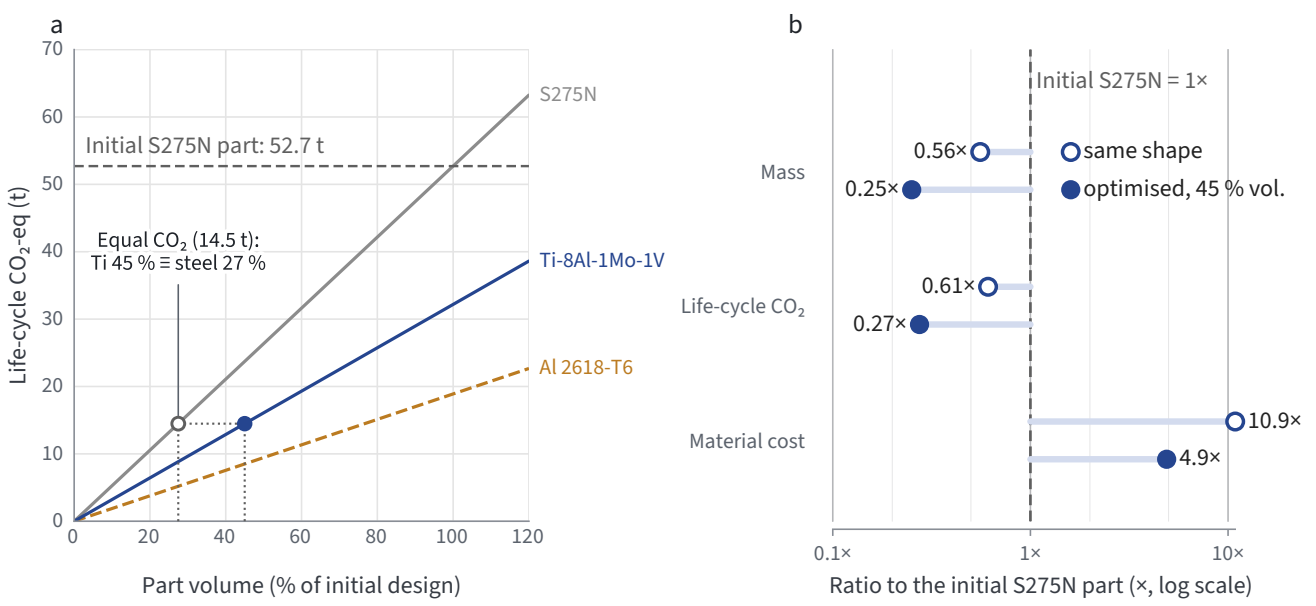
a
70
60
50
40
30
20
10
0
Life-cycle CO_2-eq (t)
Initial S275N part: 52.7 t
S275N
Equal CO_2 (14.5 t):
Ti 45 % ≡ steel 27 %
Ti-8Al-1Mo-1V
Al 2618-T6
0
20
40
60
80
100
120
Part volume (% of initial design)
b
Initial S275N = 1×
Mass
0.56×
same shape
0.25×
optimised, 45 % vol.
Life-cycle CO_2
0.61×
0.27×
Material cost
10.9×
4.9×
0.1×
1×
10×
Ratio to the initial S275N part (×, log scale)